Pulling the deeper blocks only

> Pulling only blocks 2 to 4 cut the lean° of the color embedding table to about a quarter. Holding the table off e_1 by a hard constraint went further: the spill° of the full edit fell to about the selectivity criterion, with removal° and the task as before. In both new arms, block 1, which nothing pulls, came to hold part of the concept, and most of the remaining spill comes from there. Holding the anti-subspace weight high late in training, rather than annealing it, kept the table and block 1 nearly off e_1 on the other ops and brought every `deep` run within the criterion, though removal fell somewhat.

Lean. Lightness on e_1 in the color embedding table…

Spill. The largest net drop in score on another op under the edit….

Removal. How much of `difference` the edit takes out…

The embedding-lean experiment found that every anchored run of ex-2.2.23 leans lightness onto e_1 in its color embedding table. At the embedding there is no context yet for the pull to use, so the pull settles for a token-level stand-in: darker colors are the nearest one for `difference` answers. Editing only blocks 2 to 4 of those runs removed most of `difference` with almost no spill. Editing the embedding alone spilled as much as editing every slice.

Pulling every slice but the embedding had been tried before (ex-2.2.21 and the $\tau \times \lambda_a$ sweep). But those runs also left the anti-subspace term° off the embedding, and the table leaned all the same.

Anti-subspace term. The companion to the pull…

So this experiment trains twelve runs at 400 epochs and 4 seeds:

- `deep`: the pull on blocks 2 to 4 only, and the anti-subspace term on every slice, so the first two slices are asked to stay off e_1 rather than left alone.
- `deep-clean`: the same, and after every step the e_1 component of every embedding is set to zero, so the embeddings cannot lean at all. The readout table is untied on this recipe (a separate matrix from the embedding table) and stays free.
- `whole`: the recipe of record, every slice pulled, trained again to record the alignment° at every slice and position along training.

Alignment. How far a state points along e_1…

Each run shares its seed, and so its initialization, batches and label draws, with one anchored run and one control of ex-2.2.23. The anchor weight, the pool temperature τ and both schedules are the recipe values. A second round of sixteen runs repeats `deep` and `deep-clean` with the anti-subspace weight annealed to a higher hold (E4).

Observations

- The color embedding table (E1): in `deep` the table leans° the same way as on the recipe on every seed, where the controls lean either way, but only about a quarter as far. The incidental lightness in the states at block 1 fades to about half.

  Lean. Lightness on e_1 in the color embedding table…

- Removal and spill (E2)°: the full edit removes as much in both new arms as on the recipe. Its spill° falls in `deep`, and falls to about the criterion in `deep-clean`. In both arms, editing block 1 alone removes nearly everything and gives most of the spill that is left.

  Removal. How much of `difference` the edit takes out…

  Spill. The largest net drop in score on another op under the edit….

- Where the alignment settles (E3)°: in the new arms the alignment of `difference` contexts grows at blocks 2 to 4 early in training, as the pull asks. Block 1 follows more slowly, on other ops as well as on `difference`.

  Alignment. How far a state points along e_1…

- The anti-subspace hold (E4): the higher the hold, the less the table leans and the less block 1 drifts onto e_1 on other ops. At a hold of 0.2 every `deep` run is within the criterion, but removal falls. In `deep-clean` the hold changes the spill less, but removal falls there too.

Scope

This is an exploratory study, with no preregistration and no gate. Four seeds per arm can only resolve a large change in the lean° or the spill°, against the seed range. The `whole` arm reproduces its twins in ex-2.2.23 (the task scores of each pair agree to within 0.0001), so a difference between arms at one seed comes from the change of pull, not from nondeterminism in training.

Lean. Lightness on e_1 in the color embedding table…

Spill. The largest net drop in score on another op under the edit….

Method

The anchor weight is the recipe value, and outside E4 the anti-subspace weight keeps its recipe schedule. That schedule is a multiple of the anchor weight: about 2.5× early in training, annealing to about 0.3× by about the midpoint.

The anchor term is a mean over the pulled slices, so at the same weight each of blocks 2 to 4 is pulled about 5/3 as hard in the new arms as in `whole`. The anti term is a mean over the same five slices in every arm, so it is unchanged.

The measurements are those of the embedding-lean experiment, taken on every run:

- The lightness slope of the color embedding table: how much further along e_1 the color embeddings sit for black than for white, as the slope of the e_1 component against lightness over the 216 colors. It is zero in `deep-clean`, where the table is held off e_1.
- The lightness correlation in the states: the correlation of α° with lightness at each slice, over the color positions of contexts of other ops.

Alignment. How far a state points along e_1…

- Removal$^\circ$ and spill$^\circ$ of the edit, which projects e_1 out of the state at every position on a set of slices. Both are net of the control at the same seed, and the selectivity criterion is a spill of at most 0.02.

Removal. How much of `difference` the edit takes out…

Spill. The largest net drop in score on another op under the edit….

- The task score of each op, as expected exact match$^\circ$ on held-out contexts.

Expected exact match. The task score, as the chance that a drawn answer agrees with a true one….

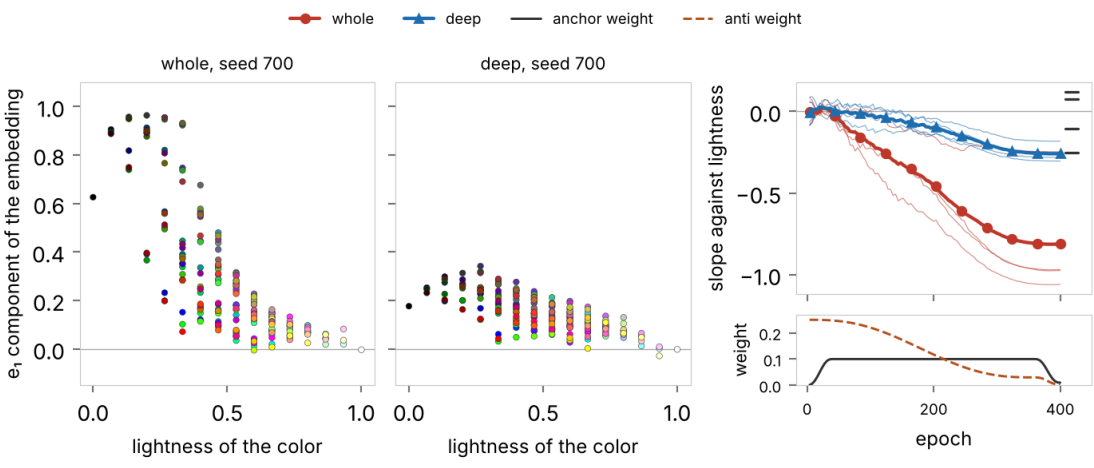
- Along training, every 4 epochs: the mean alignment α with e_1 at every slice and position of the probe contexts, for `difference` and for the other ops, and the e_1 component of every color embedding.

The color embedding table (E1)

If the lean° is the pull meeting the embedding with a stand-in, it should fade in `deep`, where the pull leaves the first two slices and the anti-subspace term° stays on them. A slope of −1 means black sits a whole unit further along e_1 than white.

Lean. Lightness on e_1 in the color embedding table…

Anti-subspace term. The companion to the pull…



The lean of the color embedding table. *Left and middle: the e_1 component of each of the 216 color embeddings against its lightness at the end of training, one run of each arm, each dot drawn in its own color. Right: the slope of that component against lightness along training, one hairline per run and the seed mean bold; the ticks at the right edge are the controls at the same seeds.* `deep-clean` *is zero throughout and not drawn. The strip below is the weight of each term, the same in both arms: the anchor weight solid and the anti-subspace weight dashed.*

arm	slope in the table	r at block 1	r at block 2
`whole`	−0.81 (−1.06 to −0.25)	−0.49 (−0.62 to −0.19)	−0.33 (−0.42 to −0.13)
`deep`	−0.26 (−0.30 to −0.18)	−0.23 (−0.24 to −0.21)	−0.15 (−0.20 to −0.10)
`deep-clean`	0.00 (0.00 to 0.00)	−0.09 (−0.15 to −0.04)	−0.09 (−0.13 to −0.07)
control	−0.04 (−0.25 to 0.12)	−0.06 (−0.43 to 0.23)	−0.08 (−0.41 to 0.24)

The lean, in the table and in the states*, seed mean and range over 4 runs. The slope is that of the e_1 component of the color embeddings against lightness; r is the correlation of α with lightness at the color positions of contexts of other ops. The controls are ex-2.2.23's at the same seeds.*

It fades to about a quarter, though not to nothing. The slope in `deep` is between −0.18 and −0.30 by seed, against about −1.0 on the recipe runs with no latch° (seed 701 of the recipe is latched on `,` and has a shallow slope too). That is about as steep as the steepest control. But the controls lean either way by seed, and every `deep` run leans the same way as the recipe, so what is left is still the pull at work, only weaker.

Latch. A syntax token embedding on e_1, which satisfies the pull at the embedding for every context at once….

The lightness in the states at block 1 fades too, to about half the recipe value, and in `deep-clean` it is smaller again. By block 2 it is small in every arm.

On the recipe the slope grows from early in training. On `deep` it stays near zero for about the first hundred epochs, then grows through the second half, while the whole table drifts a little onto e_1 (E3). The pull on block 2 reaches the table only through block 1, and that is enough to load some lightness onto e_1.

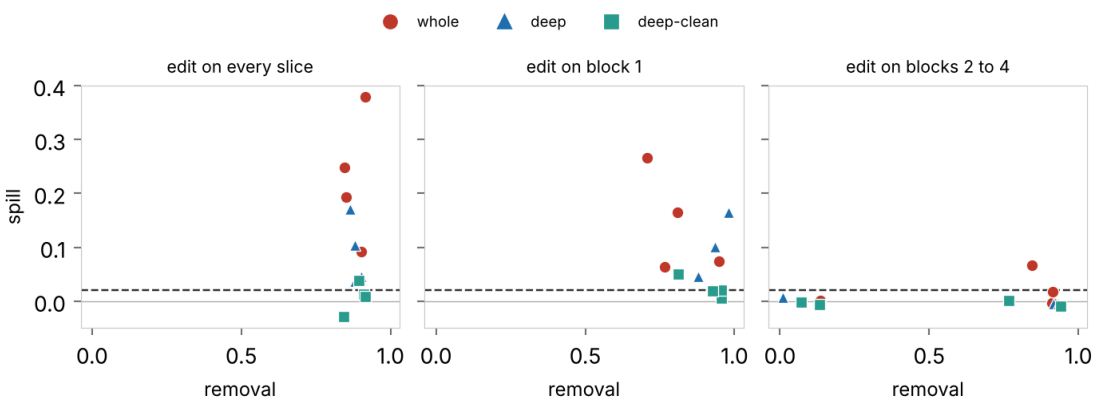
The growth also coincides with the anneal of the anti-subspace weight, which E4 changes. Part of the lean may also come from the per-slice pull being 5/3 of the recipe value.

Removal and spill (E2)

The edit on every slice is the goal, since one that has to know which slices to skip is a weaker guarantee. The main measure is its spill°, with removal° beside it.

Spill. The largest net drop in score on another op under the edit....

Removal. How much of `difference` the edit takes out...



***Spill against removal under three slice-restricted edits**, one mark per run. Removal is the net drop on `difference` as a share of the way to the target null; spill is the largest net drop on another op. The dashed rule is the selectivity criterion.*

slices edited	`whole`	`deep`	`deep-clean`
every slice: removal	0.88 (0.84 to 0.92)	0.88 (0.86 to 0.90)	0.89 (0.84 to 0.92)
embedding: removal	0.55 (0.42 to 0.74)	0.18 (0.07 to 0.30)	−0.03 (−0.08 to 0.00)
block 1: removal	0.81 (0.71 to 0.95)	0.93 (0.88 to 0.98)	0.91 (0.81 to 0.96)
blocks 2 to 4: removal	0.70 (0.14 to 0.91)	0.48 (0.01 to 0.93)	0.48 (0.07 to 0.94)
every slice: spill	0.229 (0.093 to 0.380)	0.090 (0.038 to 0.171)	0.008 (−0.029 to 0.038)
embedding: spill	0.233 (0.106 to 0.393)	0.035 (0.000 to 0.098)	−0.003 (−0.011 to 0.003)
block 1: spill	0.143 (0.064 to 0.267)	0.083 (0.020 to 0.165)	0.024 (0.006 to 0.050)
blocks 2 to 4: spill	0.021 (−0.004 to 0.067)	−0.001 (−0.005 to 0.008)	−0.004 (−0.009 to 0.001)

***Removal and spill under each slice-restricted edit**, seed mean and range over 4 runs. The `whole` runs are the same as their twins in ex-2.2.23.*

The spill falls, and removal holds. Under the edit on every slice all twelve runs remove about as much as each other. The recipe runs spill well above the criterion on every seed, and `deep` spills less on every seed but stays above it. `deep-clean` is within it on 3 of 4 runs, and the fourth spills about twice the criterion (0.038).

The task score is the same in all three arms and on the controls (about 0.57 expected exact match° averaged over ops, and no run more than 0.01 below the controls), so the new pull costs the task nothing we can see.

Expected exact match. The task score, as the chance that a drawn answer agrees with a true one....

In both new arms the concept sits at block 1 as well as later. Editing block 1 alone removes at least as much as editing every slice, though block 1 is never pulled, and it gives most of the spill that is left. In `deep` the embedding alone still spills on seeds 700 and 702, where the weaker lean° of E1 is still used downstream.

Lean. Lightness on e_1 in the color embedding table...

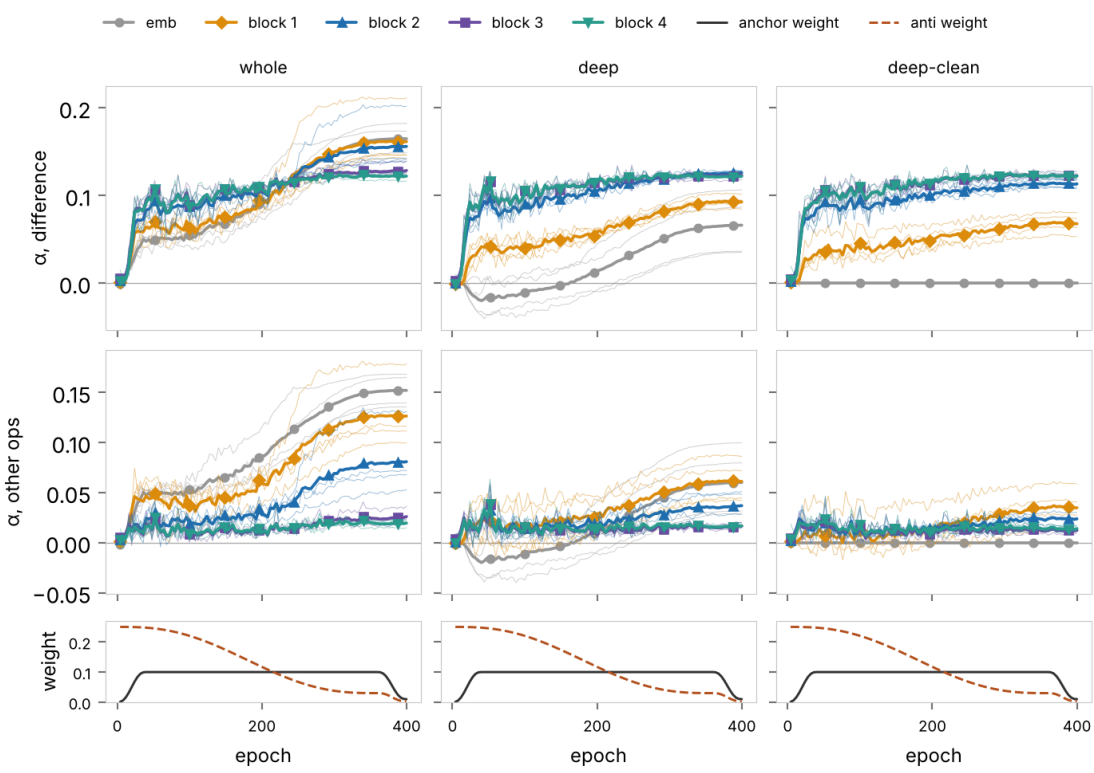
Editing only blocks 2 to 4 spills almost nothing in any arm, but its removal splits by seed. On seven of the twelve runs it removes nearly everything, and on the other five almost nothing (seeds 700 and 702 in `deep`, seeds 700 and 701 in `deep-clean`, and seed 700 in `whole`).

On those five, the answer depends on the e_1 component at block 1, and not on the e_1 component the pull put at blocks 2 to 4. Perhaps the later blocks read block 1 along e_1 and write the answer elsewhere. Or the two edits may differ in some other way that this measurement cannot tell apart. The embedding-lean experiment saw the same split on the recipe and took it as a sign that the concept was partly built from the lean. Here the split persists with the table held off e_1, so it does not need the lean.

Where the alignment settles (E3)

A concept that settles where the pull is should show up at blocks 2 to 4 on `difference` contexts and nowhere on the others. The measure is the mean of α° over the positions of a probe context, which mixes the positions where the concept sits with those where it does not, so only its shape over training and slices matters.

Alignment. How far a state points along e_1...



Mean alignment with e_1 along training, by slice. *The mean of α over the positions of the probe contexts of `difference` (top) and of the other ops (bottom), one line per slice; bold is the seed mean and the hairlines are the 4 runs. The strips under each column are the anchor weight (solid) and the anti-subspace weight (dashed), the same in every arm.*

On the recipe every slice rises together on `difference` contexts, and the embedding and block 1 rise nearly as much on the other ops. That is the lean° seen from the states: at those slices the alignment follows the tokens, whatever the op.

Lean. Lightness on e_1 in the color embedding table...

On both new arms, blocks 2 to 4 rise on `difference` contexts within the first few dozen epochs and then level off, and stay low on the other ops. Block 1 rises later and more slowly, on the other ops as well as on `difference`, though less on the others than on the recipe. So block 1 comes to meet part of the pull on block 2, and the anti term does not stop it.

Much of the rise at block 1, and nearly all of it on the other ops, comes after epoch 150, as the anti weight anneals (E4). Its rise on other ops is the spill° that E2 traced to the edit on block 1.

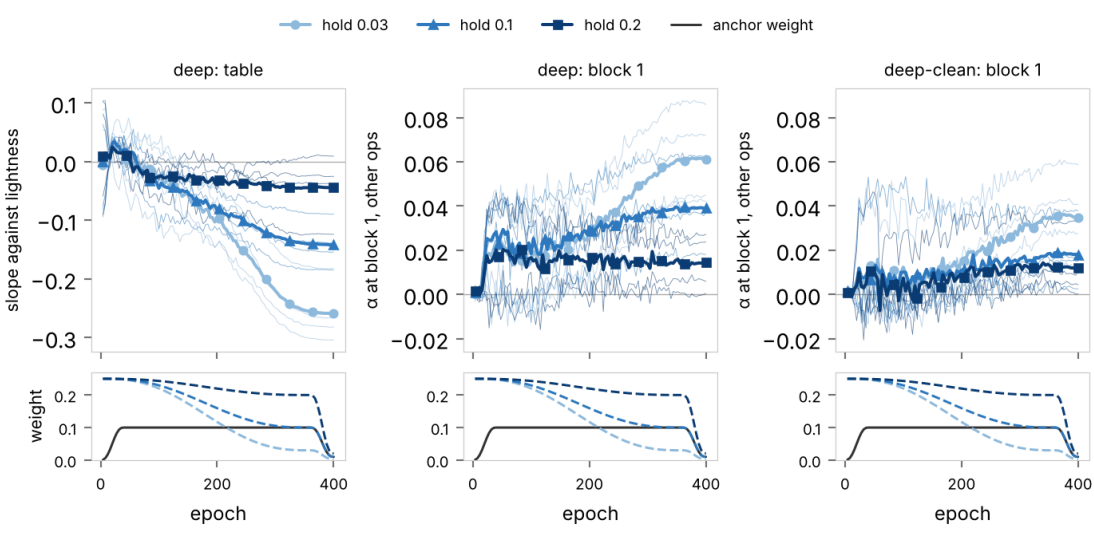
Spill. The largest net drop in score on another op under the edit....

On `deep` the embedding also drifts onto e_1 in the second half of training, by about as much on the other ops as on `difference`. That is the whole color embedding table shifting a little toward e_1, with the lightness slope of E1 on top. It does not happen in `deep-clean`, where the table is held at zero.

The anti-subspace hold (E4)

In E1 and E3 the table and block 1 moved onto e_1 mostly while the anti-subspace weight was annealing. If the anneal lets go too early, holding that weight higher should keep them off.

A second round repeats `deep` and `deep-clean` with the anti weight annealed to a hold of 0.1 (equal to the anchor weight) or 0.2, in place of the recipe hold of 0.03. The peak at the start, the anneal at the end of training, and the seeds are unchanged.



The lean and block 1 along training, by anti-subspace hold. *Left: the slope of the e_1 component of the color embeddings against lightness in `deep`. Middle and right: the mean α at block 1 over the positions of the probe contexts of the other ops. One line per hold: bold is the seed mean and the hairlines are the 4 runs. The strips below are the anchor weight (solid gray) and the anti-subspace weight at each hold (dashed).*

The higher the hold, the less the table leans° and the less block 1 drifts onto e_1 on other ops. At 0.2 the slope in `deep` is between −0.12 and 0.01 by seed. That is within the range of the controls, and no longer all one sign. Block 1 stays near its early level through training. At 0.1 both grow, but about half as much as at the recipe hold.

Lean. Lightness on e_1 in the color embedding table…

arm	anti hold	slope in the table	removal	spill	within criterion	task change
`deep`	0.03	−0.26 (−0.30 to −0.18)	0.88 (0.86 to 0.90)	0.090 (0.038 to 0.171)	0 of 4	−0.003 (−0.006 to 0.001)
`deep`	0.1	−0.14 (−0.19 to −0.09)	0.86 (0.77 to 0.92)	0.017 (−0.001 to 0.057)	3 of 4	−0.004 (−0.010 to 0.000)
`deep`	0.2	−0.04 (−0.12 to 0.01)	0.77 (0.64 to 0.86)	−0.012 (−0.027 to −0.002)	4 of 4	−0.005 (−0.018 to 0.003)
`deep-clean`	0.03	0.00 (0.00 to 0.00)	0.89 (0.84 to 0.92)	0.008 (−0.029 to 0.038)	3 of 4	−0.003 (−0.009 to −0.001)
`deep-clean`	0.1	0.00 (0.00 to 0.00)	0.79 (0.60 to 0.88)	−0.006 (−0.028 to 0.021)	3 of 4	−0.006 (−0.014 to −0.002)
`deep-clean`	0.2	0.00 (0.00 to 0.00)	0.75 (0.49 to 0.88)	0.006 (−0.031 to 0.063)	3 of 4	−0.006 (−0.010 to −0.003)

The lean, removal and spill by anti-subspace hold*, seed mean and range over 4 runs, under the edit on every slice. The task change is the mean score over ops less that of the control at the same seed. The rows at the hold of 0.03 are the `deep` and `deep-clean` runs of E1 to E3.*

In `deep` the spill° follows. The full edit is within the criterion on 3 of 4 runs at 0.1 and on 4 of 4 at 0.2, where none were at the recipe hold. But removal° falls as the hold rises, to between 0.64 and 0.86 at 0.2. With the concept kept out of block 1, less of `difference` depends on e_1 anywhere.

Spill. The largest net drop in score on another op under the edit….

Removal. How much of `difference` the edit takes out…

In `deep-clean` the hold changes the spill less, since it was already near the criterion, but removal falls about as much as in `deep`. At 0.2 one run (seed 703) spills 0.063, more than any `deep-clean` run at the recipe hold, and another (seed 701) removes only 0.49.

The task changes little. The largest drop against the control is 0.018 (seed 702 of `deep` at a hold of 0.2), against 0.009 on the runs at the recipe hold.

Discussion

Moving the pull off the first two slices did most of what the embedding-lean experiment expected of it. The color embedding table leans° much less, the lightness in the states fades, and the full edit spills° less on every seed. Holding the table off e_1 outright took the spill most of the rest of the way, with nothing lost in removal° or in the task. That fits the account that the spill comes from stand-ins at the slices where nothing contextual exists.

Lean. Lightness on e_1 in the color embedding table…

Spill. The largest net drop in score on another op under the edit….

Removal. How much of `difference` the edit takes out…

It did not keep the concept out of block 1. The pull on block 2 is cheapest to meet by having block 1 already lean toward e_1, and block 1 has some context to work with, so part of what it puts there is about `difference` and part is shared with other ops.

Keeping out the shared part is the job of the anti-subspace term°, and on its recipe schedule it does not. Block 1 and the table move onto e_1 mostly in the second half of training. By then the anti weight has annealed from about 2.5× the anchor weight to about 0.3×. Holding it at 2× the anchor weight kept both nearly off e_1 on the other ops in `deep` (E4). So on this pull the recipe anneal lets go too early.

Anti-subspace term. The companion to the pull…

The embedding-lean experiment found the pull on a color embedding to be an order of magnitude or more above what the anti term costs it. In `deep` the pull reaches the table only through block 1, and a hold of about 7× the recipe value was enough.

The higher hold also took some removal. Some of what the edit removes at the recipe hold seems to sit at block 1. With block 1 kept off e_1, the model answers `difference` with less of it on e_1.

The anti term also acts on the pulled blocks, and there the alignment° of `difference` contexts falls a little as the hold rises. So part of the cost may be the hold pressing on the pull itself.

Alignment. How far a state points along e_1…

Among the runs here, `deep` at a hold of 0.2 is the most selective. `deep` at 0.1 and `deep-clean` at the recipe hold remove more, with a little more spill.

On the seeds where the answer to `difference` depends on the e_1 component at block 1 and not on the one at the later blocks (E2), the edit on every slice is the one to keep, and the remaining spill is a question of what block 1 holds.

Four seeds show that `deep-clean` and the higher holds spill much less than the recipe, but not whether their single runs above the criterion, or the runs with low removal, are typical.

Glossary

Alignment

α

How far a state points along e_1: the cosine between the state and e_1, which is 1 on the anchor direction and 0 off it.

Anti-subspace term

The companion to the pull: the mean of α^2 over every position and slice it acts on, which asks the states as a whole to stay off e_1.

Lean

Lightness on e_1 in the color embedding table: darker colors sit further along the anchor direction, a stand-in for `difference` at the slices where nothing contextual exists.

Latch

A syntax token embedding on e_1, which satisfies the pull at the embedding for every context at once. Found by spill-by-position.

Removal

How much of `difference` the edit takes out: the net drop in its score under the edit, as a share of the way from the clean score to the target null.

Spill

The largest net drop in score on another op under the edit. The selectivity criterion is a spill of at most 0.02.

Target null

The score an ideal predictor would get on `difference` contexts if it had lost that op and answered from the remaining ones.